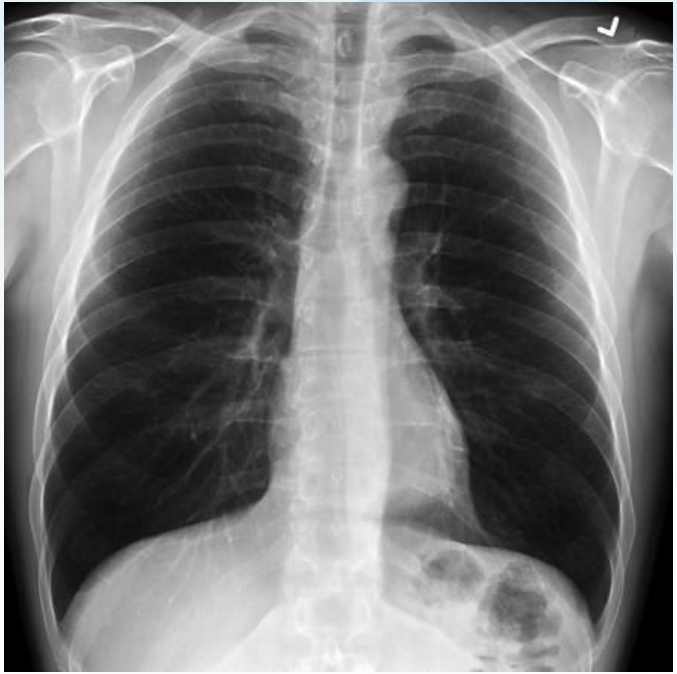


Identifying Abnormalities in Chest X-Ray Images

Data Dawgs - Ben, Ciara, Anuti, and Sahithi

Motivation and Research Question

Can image-based models accurately classify chest X-ray images into normal versus abnormal cases?

- Long wait times and radiologist shortages contribute to delays
- X-Ray demand is expected to outpace supply in coming years
- Computer-aided detection and diagnosis (CAD) in real world medical sites could be a great tool in lowering wait times

Hypothesis and Modeling Approach

Hypothesis

An EfficientNet CNN will outperform a DenseNet CNN in identifying chest X-ray abnormalities, achieving at least 75% accuracy while maintaining higher recall for abnormal cases

Model

- EfficientNet: Currently regarded as a top overall CNN
- DenseNet: Often more robust, catches subtle patterns
- Performance evaluated using AUC, probability distributions, and classification metrics

Data Overview

- A zip folder containing 4,999 chest X-Ray images and a CSV containing their results
- Sourced from the NIH (Natoinal Institutes of Health)
- Finding Labels were reclassified into Finding/No Finding

Column	Description	Potential Responses
Image Index	The filename/ID of each X-ray image	00000001_000.png
Finding Labels	The medical conditions identified in that image	Hernia, No Finding
OriginalImage, Width	The width of the original X-ray image in pixels	2894, 2500
Height]	The height of the original X-ray image in pixels	2048, 2986
OriginalImagePixelSpacing	The horizontal distance between pixels (in mm)	0.14300000000000002
y]	The vertical distance between pixels (in mm)	0.168

Analysis Plan

01

Data Cleaning

02

Reclassify Data into Normal/Abnormal

03

Train/Test Split

04

Weight classes equally

05

Create and train models

06

Evaluate and analyze success

Tricky Analysis Decisions

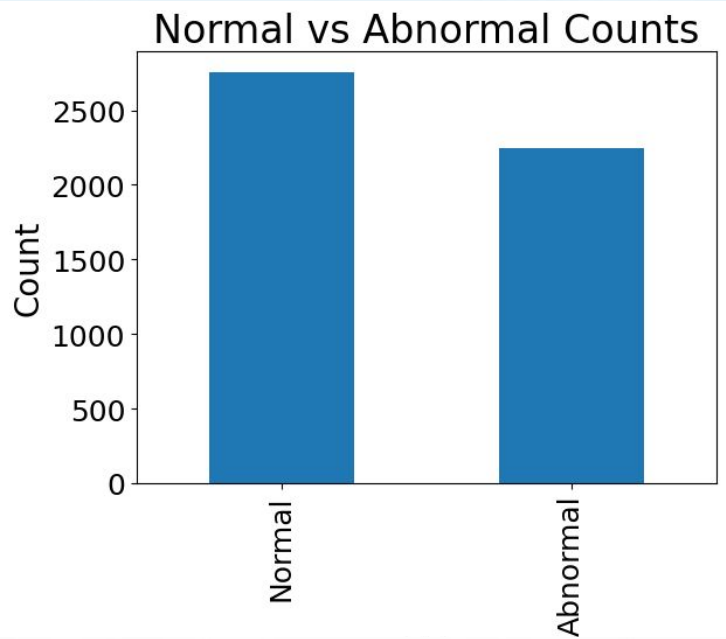
Choosing # of Epochs

- Balance between overfitting or underfitting to training data
- EfficientNet accuracy stopped improving after ~15 Epochs
- DenseNet Accuracy stopped improving after ~50 Epochs

Choosing Decision Thresholds

- Decision thresholds affect model precision and recall
- We chose the threshold that maximized the difference between true and false positive rates
- This approach best separates normal and abnormal images

Bias and Uncertainty



Potential Bias

- Classes were slightly imbalanced, leading to overclassification of X-Rays into the normal class
- This was solved by class-weighting to balance observations

Sources of Uncertainty

- We cannot be sure that the CNN models have properly learned what to look for in X-Rays

RESULTS AND CONCLUSIONS

EfficientNet

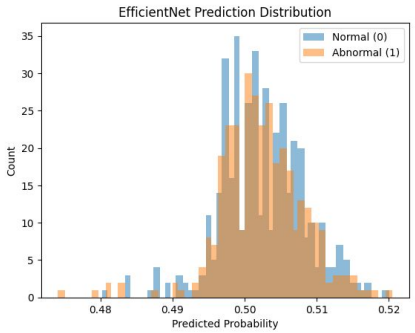
Accuracy: 0.65

Abnormal recall: 0.66

AUC: 0.69

Minimum prob: 0.056

Maximum prob: 0.936



DenseNet

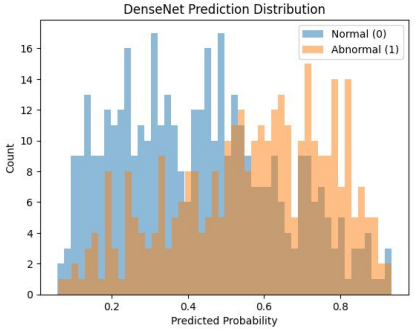
Accuracy: 0.50

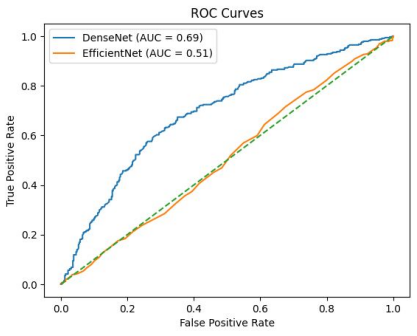
Abnormal recall: 0.64

AUC: 0.51

Minimum prob: 0.474

Maximum prob: 0.5205






```
DenseNet:
              precision    recall  f1-score   support

           0       0.70      0.65      0.67       413
           1       0.61      0.66      0.63       337

    accuracy                           0.65       750
   macro avg       0.65      0.66      0.65       750
weighted avg       0.66      0.65      0.66       750

EfficientNet:
              precision    recall  f1-score   support

           0       0.57      0.39      0.46       413
           1       0.46      0.64      0.54       337

    accuracy                           0.50       750
   macro avg       0.52      0.52      0.50       750
weighted avg       0.52      0.50      0.50       750
```

CONCLUSIONS

1. DenseNet had higher overall classification accuracy than EfficientNet
2. DenseNet had slightly better abnormal recall than EfficientNet
3. **Our hypothesis that EfficientNet will have 75% accuracy and better abnormal recall than DenseNet is not supported**

Next Steps

- Explore more CNN models like ResNet or VGG
- See if models actually learn abnormalities by using Grad-CAM and researching how abnormalities appear in X-Ray images
- Add patient metadata to see if abnormalities tend to show up in certain ages, genders, medical histories, etc…

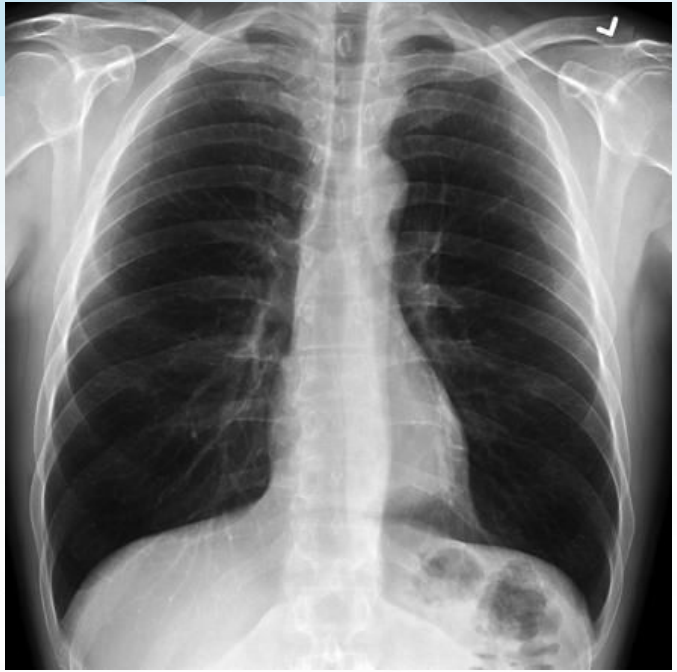
RESOURCES

[1] A. Giełczyk, A. Marciniak, M. Tarczewska, and Z. Lutowski, “Pre-processing methods in chest X-ray image classification,” *PloS one*, vol. 17, no. 4, p. e0265949, Jan. 2022, doi: https://doi.org/10.1371/journal.pone.0265949.

[2] “Box,” *nihcc.app.box.com*. https://nihcc.app.box.com/v/ChestXray-NIHCC/folder/36938765345

[3] M. Ucan, B. Kaya, O. Aygun, M. Kaya, and R. Alhajj, “Comparison of EfficientNet CNN models for multi-label chest X-ray disease diagnosis,” *PeerJ Computer Science*, vol. 11, p. e2968, Jul. 2025, doi: https://doi.org/10.7717/peerj-cs.2968.

[4] “Wait times for emergency hospitalization keep getting higher,” *Institute for Healthcare Policy & Innovation*, 2024. https://ihpi.umich.edu/news-events/news/wait-times-emergency-hospitalization-keep-getting-higher



Thank you!

Questions?